Process Transfer Plan

A-Mab Drug Substance — PTP-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Field	Value
Document ID	PTP-001
Document class	Process Transfer Plan
Title	Process Transfer Plan: A-Mab Drug Substance
Product	A-Mab
Version	2.0
Effective date	2026-09-30
Supersedes	PTP-001 Version 1.0 (2025-06-16)
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor	Novacyte Biologics
Development site	Cambridge, MA
Commercial drug substance site	Grafton, WI

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Approvals

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Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2025-06-16	Initial issue
2.0	2026-09-30	Reissued under INV-001: approval and execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
2-AB	2-aminobenzamide
A280	absorbance at 280 nm
ADCC	antibody-dependent cellular cytotoxicity
AMD	analytical method decision
AMV	analytical method validation
AU	absorbance unit
BLA	Biologics License Application
CDC	complement-dependent cytotoxicity
CE-SDS	capillary electrophoresis with sodium dodecyl sulfate
CEX	cation exchange chromatography
CH2, CH3	second and third constant domains of the antibody heavy chain
CO2	carbon dioxide
CPP	critical process parameter
CQA	critical quality attribute
CV	column volume
DoE	design of experiments
DS	drug substance
DV	diavolume
ELISA	enzyme-linked immunosorbent assay
Fc	crystallizable fragment of the antibody
FcγRIIIa	Fcγ receptor IIIa
GPP	general process parameter
HCP	host cell protein
HILIC	hydrophilic interaction liquid chromatography
HMW	high molecular weight species
HPLC	high-performance liquid chromatography
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
IgG1	immunoglobulin G, subclass 1
INV	investigation report
kLa	volumetric mass transfer coefficient

Abbreviation	Meaning
KPP	key process parameter
LMW	low molecular weight species
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
NTU	nephelometric turbidity unit
OD280	optical density at 280 nm
PA	Protein A
PAR	proven acceptable range
PCMP	Process Characterization Master Plan
PCMR	Process Characterization Master Report
pCO2	CO2 partial pressure
PCP	Process Characterization Plan
PCR	Process Characterization Report
PPQ	process performance qualification
PTP	Process Transfer Plan
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
RA	risk assessment
rcf	relative centrifugal force
SEC	size exclusion chromatography
SOP	standard operating procedure
TCID50	median tissue culture infectious dose
UF/DF	ultrafiltration and diafiltration
UPLC	ultra-performance liquid chromatography
UV	ultraviolet
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukemia virus

1 Purpose and scope

This plan governs the transfer of the A-Mab drug substance manufacturing process from the development site of Novacyte Biologics in Cambridge, MA to its commercial drug substance site in Grafton, WI. The transfer covers the manufacturing process, the analytical methods that test the intermediates and the drug substance, and the control strategy that connects them. Section 2 describes the process, and Section 3 names the work that each site holds.

This plan is the parent document of the A-Mab process characterization package. The package has 22 documents, including this plan, and Table 1.1 lists the others. The risk assessment RA-001 sets the scope of each study. Each Process Characterization Plan (PCP-003 to PCP-010) commits the studies of its step, and each paired Process Characterization Report (PCR-003 to PCR-010) records what they found. The master plan PCMP-001 sets the framework that the step plans apply, and the master report PCMR-001 consolidates the reports. Together these documents justify the operating ranges and the control strategy that the receiving site will run. The package also holds two investigation reports of Quality Assurance. The investigation INV-001 is the reason for this revision (Section 1.1), and the investigation INV-002 concerns the adequacy of the response-surface models in the step reports.

Table 1.1: The other documents of the A-Mab characterization package.

Document ID	Document class	Subject
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)

Document ID	Document class	Subject
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance
INV-001	Investigation Report	Acceptance criteria for proven-acceptable-range analysis set after data review
INV-002	Investigation Report	Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0

Three activities are outside the scope of this plan. Drug product manufacture, filling and finishing are transferred under a separate plan. Cell bank manufacture and testing are governed by their own records. The execution of the process performance qualification (PPQ) campaign is also outside this plan. This plan sets the conditions the campaign must meet before it may start and the strategy it will follow (Section 7). The campaign will be executed under its own protocol and reported separately.

The transfer follows ICH Q10, which places technology transfer within the pharmaceutical quality system (International Council for Harmonisation 2008), and the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). In that lifecycle, process characterization belongs to Stage 1 (process design). Stage 2 (process qualification) covers the qualification of the facility, equipment and utilities and the process performance qualification. The enhanced development approach of ICH Q8 and ICH Q11 governs how the design space and the control strategy are defined (International Council for Harmonisation 2009, 2012). ICH Q9 governs the risk assessment that scopes the studies (International Council for Harmonisation 2023b), and ICH Q5A governs the evaluation of viral safety (International Council for Harmonisation 2023a). The process model, the quality attribute framework and the risk method follow the published A-Mab case study (CMC Biotech Working Group 2009).

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-06-16, before the risk assessment RA-001 (2025-07-14), the master plan PCMP-001 (2025-08-04) and the first characterization study, the qualification of the production bioreactor scale-down model (S-BRX-SDM), which started on 2025-08-11. Version 2.0 replaces it under corrective action CA3 of the QA investigation INV-001, which reissues this plan, the risk assessment RA-001, the step reports PCR-003 to PCR-010 and the master report PCMR-001 at Version 2.0. Quality Assurance opened the investigation on 2026-08-03, after a review of the master report PCMR-001 Version 1.0 for readiness for the Biologics License Application (BLA), and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the proven acceptable range (PAR) analysis had been set after the data were seen and were in no approved protocol (INV-001 finding 1). This plan set no step criteria. This revision makes the following changes:

1. The title block of Version 1.0 carried 2026-07-24, the date on which the Version 1.0 package was compiled, in place of the date on which this plan was approved (INV-001 finding 5). The revision history now gives the approval date. Section 10 records the approval date of each step plan and the execution window of the studies that Version 1.0 reported (Table 10.2).
2. Where this plan states what an intermediate is judged against, it now refers to the intermediate acceptance criteria of the master plan PCMP-001 Version 2.0 §7.3. This applies to the comparison of the sites (Section 5), the control strategy (Section 6.3) and the qualification campaign (Section 7). These criteria are derived a priori from the drug substance criteria and the platform worst case of each downstream step. Version 1.0 referred to the limit that each characterization report would establish for its own step. The investigation found that the step limits had been set after the data were seen and were in no approved protocol (finding 1), that the aggregate and host cell protein limits carried the drug substance limit back through the nominal clearance of this campaign and divided it by a margin that differed by step (finding 2), and that the bioreactor limits for the glycans and acidic variants were computed from the capability data they were then used to judge (finding 3). The viral clearance claims of low-pH viral inactivation, anion exchange and virus filtration now refer to PCMP-001 Version 2.0 §7.4.
3. Where this plan describes the qualification of the scale-down models (Section 4, Section 5 and Table 8.1), it now refers to the offset rule of PCMP-001 Version 2.0 §7.1. Under that rule a model is qualified for each output by its offset, the scale-down mean minus the manufacturing-scale mean, and not by a test of equivalence. For the outputs that the qualification compares at both scales, every judgement against a criterion is stated at manufacturing scale. The manufacturing-scale reference is the development train at the sending site (Section 5). Version 1.0 said in Table 4.1 that the models are qualified against the commercial train before any study is run, which the order of Table 10.1 rules out. Its gap action recorded that comparison in the step plans, which were approved before their studies. The commercial train is now compared with the development train

on the engineering batches, and the comparison is recorded in the engineering batch reports (Table 8.1).

4. The quality attributes in Table 2.2 now include fragments (low molecular weight species). The sponsor set the criterion for fragments, measured by non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS), when fragments joined the CQA register of PCMP-001 Version 2.0. The A-Mab case study lists fragmentation among the typical quality attributes of an antibody but gives it no limit (PCMP-001 §3). That register assigns fragments to harvest, so the description of harvest in Table 2.1 and the basis on which the harvest gap is accepted (Section 8) are restated. The nrCE-SDS method (SOP-3015, AMV-3020) is added to the methods to transfer (Section 6.2 and Table 8.1), following the analytical method decision AMD-001 of 2026-03-20 as PCMP-001 Version 2.0 §7.8 sets it. This answers INV-001 finding 6, that size exclusion chromatography (SEC) reads fragments high in samples that still carry host cell protein.
5. The procedures and method validations in Table 6.2 now also include the product titre method by Protein A affinity HPLC (SOP-3016, AMV-3021), because the titre sets the protein load of a capture run, and the method for aggregate by SEC-HPLC (AMV-3221). Table 6.3 lists the methods to be transferred with their qualified performance. Beside the methods of the Version 1.0 table, it now includes the charge variant method by icIEF (AMV-3013), the residual DNA method by qPCR (AMV-3014) and the product titre method (AMV-3021). It no longer includes the two infectivity titre assays (AMV-3017 and AMV-3018), which stay with the virus spiking studies at small scale (Section 6.2).
6. Section 6.1 now carries the parameter table as it stands at this revision (Table 6.1), with the set-point, NOR, characterized range and class of every parameter. It includes the set-points and NORs that the Protein A report PCR-005 and the low-pH viral inactivation report PCR-006 changed at Version 2.0, and its footnotes give the value the studies were planned against.
7. Table 1.1 lists the two investigation reports, INV-001 and INV-002, beside the plans and reports of the package.
8. Statements of Version 1.0 about the product, the sites and the scale-down models that were wrong, or that the record now contradicts, are corrected. Version 1.0 linked both glycan attributes to ADCC, but galactose content raises the complement-dependent cytotoxicity of A-Mab, not its ADCC, and aggregate also forms at low pH (Section 2). Protein A and cation exchange bind the antibody, and only anion exchange binds the impurities while the antibody flows through (Section 2). The CQA register assigns each viral clearance requirement to one step, where Version 1.0 said that the requirements belong to no single step (Section 2). The step roles in Table 2.1 are those of the register at this revision. The statement that a common quality system is a condition ICH Q10 places on the transfer is removed, and the handovers that Quality Assurance approves are the deliverables of Table 10.1 (Section 3). Bed height, linear velocity and the load per membrane area are held constant across scale, and the virus filter and the UF/DF step run at a set pressure (Section 4 and Section 5). A narrower distribution of a parameter does not invalidate its range because every NOR lies within its characterized range, as Table 6.1 shows (Section 4). The other elements of Table 4.1 differ between the sites in ways that can affect the process, not only in documentation, and method transfer needs laboratory work as buffer prepara-

tion does (Section 4). Version 1.0 judged comparability against the acceptance criteria alone. As ICH Q5E expects, the engineering batches are now also compared with the development-train batches (Section 5).

9. Statements of Version 1.0 about the transfer elements, the qualification campaign and the gaps are corrected in the same way. A method transfer succeeds on accuracy within the limits the transfer protocol sets, not within a qualified range, and the protocol also sets the allowed difference between the sites (Section 6.2). Release testing does not test residual DNA or viral clearance against their criteria (Section 6.3). XMuLV clearance rests on three claimed steps and MVM clearance on two, where Version 1.0 said that viral safety rests on three independent steps (Section 6.3). Stage 2 of the lifecycle also covers the qualification of the facility, equipment and utilities (Section 1 and Section 7). The qualification protocol, not the step plans, sets the sampling plan of the campaign, and post-clarification turbidity is measured, not set (Section 7). In Table 8.1 the first two gaps no longer set the schedule, which Section 10 gives, and the gaps on harvest and UF/DF and on viral clearance are managed, not closed.
10. The engineering batches are run under the master batch record, which follows PCMR-001, as the deliverables of Version 1.0 have it (Table 10.1). Version 1.0 also said that the engineering batches could run against the current process description (Section 6.3 and Table 8.1) and that operator training did not wait for PCMR-001 (Section 10). These statements now follow the order of Table 10.1.

The scope of the transfer, the site analysis, the transfer elements, the gap analysis and the strategy for process performance qualification are otherwise as Version 1.0 stated them.

2 Product and process description

A-Mab is a humanized IgG1 monoclonal antibody. Its primary mechanism of action is antibody-dependent cellular cytotoxicity (ADCC). An antibody without core fucose on its N-linked glycan binds the Fcγ receptor IIIa (FcγRIIIa) of natural killer cells more strongly, so the ADCC activity of A-Mab rises with afucosylation (CMC Biotech Working Group 2009, 38). Galactose content does not change the ADCC of A-Mab, but its complement-dependent cytotoxicity (CDC) rises with galactose content (CMC Biotech Working Group 2009, 38). Afucosylation and galactosylation are therefore quality attributes of the drug substance. The glycan is formed in the production bioreactor. None of the platform purification steps separates glycosylation variants under the conditions used here (CMC Biotech Working Group 2009). Protein A binds the Fc region at the interface of the CH2 and CH3 domains. The glycan takes no part in that binding, so Protein A captures every glycan variant alike. Fucose, galactose and mannose carry no charge, so the two ion exchange steps, which separate by charge, do not resolve these variants either. Control of the glycan attributes therefore sits in cell culture. For this reason the transfer of the cell culture step carries more risk to product quality than the transfer of any single purification step.

The drug substance is made at a production bioreactor working volume of 15,000 L. It leaves the final formulation step at a target concentration of 75 g/L. The train has 8 unit operations, numbered Step 3 to Step 10 (Table 2.1).

Table 2.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate attributes
4	Harvest and Clarification	Primary recovery and clarification, where fragments form during the harvest hold
5	Protein A Chromatography	Capture; principal HCP and DNA clearance; source of leached Protein A
6	Low-pH Viral Inactivation	Low-pH hold; inactivates enveloped virus (XMuLV credit)
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; clears HCP, DNA and virus (XMuLV and MVM credit)
9	Small-Virus Retentive Filtration	Dedicated small-virus removal by size; principal MVM clearance (MVM and XMuLV credit)
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The purification sequence is the platform sequence of the earlier humanized IgG1 products of Novacyte Biologics, X-Mab, Y-Mab and Z-Mab. Capture on Protein A is followed by a low-pH hold, cation exchange, anion exchange, small virus retentive filtration, and ultrafiltration with diafiltration (UF/DF). The low-pH hold, the anion exchange step and the virus filter each carry a viral clearance claim. The three claims are modular in the sense of ICH Q5A (International Council for Harmonisation 2023a). Each step is studied and claimed on its own, and no single step report claims the cumulative clearance of the train. The master report PCMR-001 states the cumulative clearance.

Table 2.2 lists the 11 quality attributes that the transfer must preserve. Criticality is a continuum, not a binary split into critical and non-critical, as in the case study (CMC Biotech Working Group 2009) and the risk principles of ICH Q9 (International Council for Harmonisation 2023b). The attributes ranked high and very high drive the scope of the studies.

Afucosylation, galactosylation and high mannose depend on how far the cell processes the glycan in the Golgi. That processing changes with the nutrient supply and with the dissolved gas concentrations in the bioreactor. Acidic charge variants accumulate through the deamidation of asparagine residues. The rate of deamidation rises with pH and temperature, and the extent rises with the time the product spends at those conditions. Aggregate forms in the culture and again at low pH, during the elution from Protein A and the viral inactivation hold. Cation exchange reduces it, because the resin binds the high molecular weight species more strongly than the monomer, and these species stay on the column when the pool is cut. The CQA register of PCMP-001 Version 2.0 assigns fragments to harvest, because proteases released from lysed cells clip the antibody in culture and during the harvest hold (PCMP-001 §3). Fragments are judged at the downstream steps. Host cell protein, residual DNA and leached Protein A are removed by the chromatography steps. Protein A and cation exchange bind the antibody and elute it away from most of these impurities. Anion exchange works the other way, binding the impurities while the antibody flows through. The two viral clearance requirements are cumulative. The register assigns the XMuLV requirement to low-pH viral inactivation and the MVM requirement to anion exchange, but each requirement is met by the sum of the claims of several steps (Section 6.3).

Table 2.2: Drug substance quality attributes, acceptance criteria and assigned criticality.

Category	CQA	Acceptance	Criticality level	Tool #1 score
Glycosylation	Afucosylation	2–13 % of glycans	H	60
	Galactosylation (total %Gal)	10–40 % (G1+G2 equiv.)	H	48
	High Mannose	3–10 % of glycans	VH	80
Size variant	Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60
	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12
Charge variant	Acidic charge variants (deamidation)	18–40 % (CEX/icIEF)	VL	4
Process impurity	Host Cell Protein (HCP)	≤ 100 ng/mg	M-H	36
	Residual DNA	≤ 0.001 ng/dose	VL	6
	Leached Protein A	≤ 5 ppm	L-M	16
Viral safety	Viral clearance — XMuLV (enveloped)	≥ 16.7 log10 (cumulative)	VH	20
	Viral clearance — MVM (parvovirus)	≥ 8.6 log10 (cumulative)	VH	20

Acceptance criteria from the A-Mab case study, except the criterion for fragments, which the sponsor set when fragments joined the register at Version 2.0 (PCMP-001 §3). VL, very low. L, low. M, moderate. H, high. VH, very high. A level such as M-H lies between the two levels it names. Tool #1, the criticality score of the case study, the product of impact and uncertainty. The two viral clearance attributes carry very high criticality by rule, because a viral contamination would directly harm the patient (PCMP-001 §3).

3 Sending and receiving sites

The sending site is the development site of Novacyte Biologics in Cambridge, MA. The receiving site is the commercial drug substance site of Novacyte Biologics in Grafton, WI. Both sites belong to the same sponsor and work under one pharmaceutical quality system and one change control procedure. A change to a set-point, a range or an analytical method during the transfer is therefore assessed once and applied at both sites.

The development site keeps the qualified scale-down models, the execution and the statistical analysis of the characterization studies, and the authorship of the characterization plans and reports. It also keeps the process description and the justification of the operating ranges until those are approved. The receiving site takes ownership of commercial manufacture. That ownership covers the master batch record, the operation of the equipment train at commercial scale, the engineering and qualification batches, and the routine testing that supports batch release.

The division between the sites is one of work, not of accountability. Both sites review and approve the process description, and Quality Assurance at both sites approves each handover, that is each deliverable in Table 10.1. Where a characterization study supports a range that the receiving site cannot hold, the two sites resolve it together before the range is written into a batch record.

4 Site equivalency analysis

Equivalency was assessed for each element that can carry a process difference between the two sites (Table 4.1). The assessment is prospective. Each row therefore states what is compared and how a difference is managed, not a measured outcome.

Table 4.1: Site equivalency assessment for the elements that can carry a process difference.

Element	Sending site	Receiving site	How the difference is managed
Equipment train and scale	Scale-down models of the unit operations, and the development train	Commercial equipment train at full working volume	Each model is qualified for each output by its offset from the batches of the development train (PCMP-001 Version 2.0 §7.1, Section 5). The commercial train is compared with the development train on the engineering batches (Table 8.1)
Process control capability	Laboratory controllers, with manual intervention available	Automated control with recorded sequences and alarms	Control capability is assessed parameter by parameter against the NOR of each parameter (Table 6.1)
Utilities and buffer supply	Buffers prepared for each experiment at laboratory scale	Central buffer preparation and hold, with water and clean steam from the site utilities	Buffer identity, concentration, pH and conductivity are controlled to one specification at both sites, and buffer hold times are qualified at the receiving site
Analytical capability	Methods qualified and in routine use for characterization	Methods to be transferred and requalified before comparability material is tested	Method transfer and requalification (Section 6.2)

Element	Sending site	Receiving site	How the difference is managed
Raw materials	Development lots released against the common specifications	Commercial lots under supplier qualification	Specifications are common, and a change of vendor is assessed through the shared change control procedure
Personnel and training	Study scientists trained on the scale-down models	Operators trained on the master batch record	Training is completed and verified before the engineering batches are run

The largest difference between the sites is scale. The receiving site runs the production bioreactor at 15,000 L, while the characterization studies run at the scale-down volumes qualified under SOP-1001 (Qualification of Scale-Down Models). In cell culture, scale acts through mixing and through gas transfer. At constant power per unit volume, mixing time rises with vessel volume. A bolus of feed therefore takes longer to disperse, and cells near the addition point spend longer at a high local concentration. Carbon dioxide is stripped less efficiently as the liquid column deepens, because a sparge bubble that rises through more liquid comes closer to equilibrium with the culture before it leaves. Oxygen crosses the same interface in the other direction, from the bubble into the culture. Carbon dioxide is far more soluble in the culture than oxygen. Its removal is therefore limited by the volume of gas that passes through the culture, while the oxygen supply is limited by the volumetric mass transfer coefficient (kLa) of the vessel. For this reason oxygen supply and carbon dioxide removal are matched separately when a bioreactor model is qualified.

In the purification steps, scale acts through column packing and through pressure. A chromatography column is scaled by holding the bed height and the linear velocity constant. This keeps the residence time of the mobile phase, but not the wall support that stabilizes a narrow bed. A wide commercial column can therefore lose plate count at a pressure drop that the model tolerates. A filter is scaled by membrane area. The virus filter runs at a set filtration pressure, and the UF/DF step at a set transmembrane pressure. In UF/DF the protein concentrates at the membrane surface (concentration polarization), and in the virus filter aggregates plug the pores. At a set pressure, a feed that carries more protein or more aggregate therefore gives a lower flux and, in the virus filter, a lower capacity. Section 5 describes how a model is qualified against these differences.

The second difference is process control. Automated control at the receiving site holds a parameter closer to its set-point than manual control does, and so narrows the distribution that the commercial process will deliver. A narrower distribution does not invalidate a range, because every NOR lies within the characterized range of its parameter (Table 6.1). The reverse case matters more. Where the receiving site cannot hold a parameter inside the NOR that the characterization assumes, either the range must be widened by further study or the equipment must be changed.

The remaining elements in Table 4.1 differ between the sites in ways that can affect the process, and each is managed as the table states. Central preparation adds a hold before use, during which a buffer can take up carbon dioxide, which lowers its pH, and its bioburden can grow. Commercial raw material lots are a known source of variation in culture performance and glycosylation. The transferred methods are run by other analysts on other instruments. Buffer preparation and method transfer both need laboratory work at the receiving site, a study of buffer hold times and a transfer exercise for each method, and both are listed in Section 8.

5 Scale-down model and comparability strategy

The characterization studies do not run at commercial scale. They run on scale-down models, and every conclusion drawn from them reaches the commercial process through the qualification of those models. Qualification is performed under SOP-1001 (Qualification of Scale-Down Models). The lifecycle guidance expects a manufacturer to understand how far a laboratory model represents the commercial process (U.S. Food and Drug Administration 2011).

A model is set up in two parts. Parameters that do not depend on scale, such as pH, temperature and dissolved oxygen, are run at the values proposed for commercial manufacture. Parameters that depend on scale are set so that the model reproduces the performance of the commercial train. In the production bioreactor these are agitation and gas flow, which are set to match the oxygen transfer and the carbon dioxide stripping of the commercial vessel. For the chromatography steps the model keeps the bed height and the linear velocity of the manufacturing column, and so its residence time. Load, wash and elution volumes are normalized to column volumes, and column efficiency is checked by plate count and peak asymmetry. The virus filter model uses the same filter type at the same filtration pressure and load per membrane area. The UF/DF model scales membrane loading and cross flow by membrane area at the same transmembrane pressure. The model is then compared with manufacturing scale on the outputs the step controls. For a chromatography step those outputs are step yield, pool volume and the impurity levels in the pool. For the production bioreactor they are viable cell concentration, titre and the product quality attributes the step forms.

Each model is qualified for each output separately, by the offset rule of the master plan PCMP-001 Version 2.0 §7.1. The offset of an output is the mean of the scale-down runs at set-point minus the mean of the manufacturing-scale batches, with its two-sided 90 % confidence interval. These batches come from the development train, the manufacturing-scale train at the sending site whose batches are the manufacturing-scale reference of each step’s qualification (PCMP-001 §7.1). A model may differ in mean from manufacturing scale, provided the offset is known precisely enough. For an output compared at both scales, every judgement of a scale-down result against a criterion therefore subtracts the offset first, so ranges and conclusions are stated at manufacturing scale. A model is qualified for an output when the conclusion over the NOR is the same with the offset at either end of its interval.

The qualification runs are made, and the offset of each output is estimated, before the first characterization run of the step. For a step with a response-surface design, the decision whether the model is qualified for an output is made at analysis, because

the conclusion over the NOR comes from that model. The qualification does not compare fragments or viral clearance at both scales. Their offset is unknown, not zero. Their representativeness rests on the mechanism of each step for fragments, such as clipping by proteases released from lysed cells during the harvest hold, and on the virus spiking studies for viral clearance (PCMP-001 §7.1).

Table 5.1 gives the scope of the characterization by unit operation.

Table 5.1: Characterization scope per unit operation and the documents that cover it.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

Across the train, 37 parameters are in scope. Of these, 22 are studied in multivariate designs and 15 one factor at a time. The risk assessment RA-001 assigned no parameter of harvest (Step 4) or UF/DF (Step 10) to multivariate study, so neither step carries a designed experiment. Their ranges rest on univariate work and on platform experience, and no interaction between two of their parameters is estimated (Section 8).

Comparability between the two sites is demonstrated on material, not on documents. The receiving site will run engineering batches on the commercial train before the qualification campaign starts. Each batch will be sampled at every step and tested by the transferred methods. Under ICH Q5E a change of manufacturing site is judged by comparing the quality attributes before and after the change, and meeting the acceptance criteria alone is generally not enough (International Council for Harmonisation 2004). The drug substance from the engineering batches will therefore be compared with the drug substance of the development-train batches on the attributes of Table 2.2 that a batch test measures. Each batch must also meet the acceptance criteria in Table 2.2, except residual DNA and viral clearance, whose criteria are met through clearance (Section 6.3). Each intermediate will be compared with its intermediate acceptance criterion from PCMP-001 §7.3, where one is defined.

Prior knowledge from the earlier humanized IgG1 products supports the expectation that the platform purification steps behave the same way at both scales (CMC Biotech Working Group 2009). The comparison of the engineering batches with the development-train batches tests that expectation.

6 Transfer of the process, methods and control strategy

The transfer is delivered in three parts. Section 6.1 covers the manufacturing process, Section 6.2 the analytical methods and Section 6.3 the control strategy.

6.1 The manufacturing process

The process description is the source document of the transfer. It states the sequence of unit operations, the materials, the set-point of every parameter and the range within which each parameter may move. The characterization studies confirm the set-points and ranges, and in some cases narrow or move them. The confirmed values are issued as a revision of the process description before the master batch record is drafted.

Table 6.1 gives every process parameter with its set-point, NOR, characterized range and class as they stand at this revision. The class is the one assigned in the report of the step under the parameter classification procedure SOP-4001. The set-point and the NOR are those in force. Some of them were changed at Version 2.0 by the report of their step, after the studies. For Protein A, the NOR of protein load was narrowed (PCR-005). For low-pH viral inactivation, the NOR and the set-point of inactivation pH were moved, the NOR of hold time was narrowed and its set-point moved, and the NOR of temperature was narrowed (PCR-006). The report of the step gives the reason for each change.

Table 6.1: Process parameters by step: set-point, ranges and classification.

Step	Parameter	Unit	Set-point	NOR	Characterized range	Class
Production bioreactor	Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP
	Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP
	CO2 partial pressure (pCO2)	mmHg	100	70–130	40–160	WC-CPP
	Osmolality	mOsm/kg	400	375–425	360–440	WC-CPP
	Culture duration	days	17	16–18	15–19	WC-CPP
	Dissolved oxygen[a)]	% air saturation	50	40–60	30–70	KPP
	Initial viable cell conc.[a)]	10^6 cells/mL	1	0.8–1.2	0.5–1.5	KPP
	Basal medium concentration[a)]	× standard	1.2	1–1.4	0.8–1.6	GPP
	Nutrient feed-1 volume[a)]	% of working volume	12	10–14	9–15	KPP
Harvest	Centrifugation (rcf)[a)]	g	9000	8000–10000	6000–12000	KPP
	Depth filter load[a)]	L/m2	120	100–140	80–160	KPP
	Post-clarification turbidity[a)]	NTU	5	1–10	1–20	GPP
Protein A	Protein load	g/L resin	30	20–39[c1)]	10–50	WC-CPP
	Elution buffer pH	pH	3.55	3.4–3.7	3.2–3.9	WC-CPP
	Load flow rate	cm/hr	200	150–250	100–300	KPP
	End of pool collect	CV	2.6	2.3–2.9	2–3.2	KPP
	Operating temperature[a)]	°C	22	18–26	15–30	GPP
	Bed height[a)]	cm	20	15–25	10–30	GPP
Low-pH viral inactivation	Inactivation pH	pH	3.4[c3)]	3.3–3.5[c2)]	3.2–4	CPP
	Hold time	min	110[c5)]	100–120[c4)]	60–180	WC-CPP
	Temperature	°C	21	20–23[c6)]	15–25	WC-CPP
	A-Mab concentration[a)]	g/L	20	as characterized[b)]	5–35	GPP
Cation exchange	Protein load	g/L resin	25	18–32	10–40	WC-CPP
	Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP
	Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP
	Elution stop collect	AU (OD280, descending)	1	0.7–1.3	0.5–1.5	WC-CPP
	Elution flow rate[a)]	cm/hr	200	150–250	100–300	GPP
Anion exchange	Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP
	Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP
	Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP
	Protein load	g/L resin	200	120–280	50–300	WC-CPP
	Operating flow rate[a)]	cm/hr	300	200–400	150–450	WC-CPP
Virus filtration	Filtration volume (load)	L/m2	90	50–105	50–140	WC-CPP
	Filtration pressure	psi	13	10–20	8–30	WC-CPP
UF/DF	Number of diavolumes[a)]	DV	8	7–9	7–10	KPP
	Transmembrane pressure[a)]	psi	12	10–14	10–15	KPP
	Final DS concentration[a)]	g/L	75	70–80	65–85	KPP

[a)] Studied one factor at a time. Every other parameter was studied in the designed experiments of its step.
[b)] The normal operating range equals the characterized range.
[c1)] Narrowed at Version 2.0 from 20–40 g/L resin, the NOR as characterized (PCR-005).
[c2)] Moved at Version 2.0 from pH 3.4–3.6, the NOR as characterized (PCR-006).
[c3)] Moved at Version 2.0 from pH 3.5, the set-point in the characterization plan (PCR-006).
[c4)] Narrowed at Version 2.0 from 60–120 min, the NOR as characterized (PCR-006).
[c5)] Moved at Version 2.0 from 90 min, the set-point in the characterization plan (PCR-006).
[c6)] Narrowed at Version 2.0 from 19–23 °C, the NOR as characterized (PCR-006).

The receiving site will draft the master batch record from the approved process description. The batch record must carry every parameter classified as a critical or a well-controlled critical process parameter, 21 in all, with its set-point and its NOR. It must also carry the in-process controls and the sampling points that support them. Where a parameter is monitored but not held to a limit, the batch record should record the measured value without an acceptance criterion. A change in the value is then visible without generating a deviation.

Table 6.2 lists the procedures and method validations referenced in this plan, which are the documents that support the transfer. The procedures for the operation of each unit operation will be reissued in the document system of the receiving site before the engineering batches are run. Three procedures stay with development, because the work they govern stays at the sending site. SOP-1001 governs scale-down model qualification, SOP-1002 the design and statistical analysis of the characterization studies, and SOP-4001 parameter classification.

Table 6.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2003	Operation of the Production Bioreactor (Fed-Batch)
	SOP-2004	Cell-Culture Media and Feed Preparation
	SOP-2005	Harvest by Continuous Disk-Stack Centrifugation
	SOP-2006	Clarification by Depth and Sterile Filtration
	SOP-2007	Operation of the Protein A Capture Chromatography Step
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-2009	Operation of the Low-pH Viral Inactivation Step
	SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step
	SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step
	SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step
	SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step
	SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC
	SOP-3011	Size-Variant Analysis by SEC-HPLC
	SOP-3012	Host-Cell-Protein Quantitation by ELISA
	SOP-3013	Charge-Variant Analysis by icIEF
	SOP-3014	Residual Host-Cell DNA by qPCR
	SOP-3015	Purity by Non-Reduced CE-SDS
	SOP-3016	Product Titre by Protein A Affinity HPLC
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3013	Charge Variants (icIEF)
	AMV-3014	Residual DNA (qPCR)
	AMV-3015	Turbidity by Nephelometry (NTU)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3017	XMuLV Infectivity Titre (TCID50)
	AMV-3018	MVM Infectivity Titre (TCID50/qPCR)
	AMV-3019	Protein Concentration by A280 (UV)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3021	Product Titre (Protein A Affinity HPLC)
	AMV-3221	Aggregate by SEC-HPLC (verification-qualified)

Handover of the manufacturing process is complete when the process description, the master batch record, the bill of materials and the operator training records have been approved at the receiving site. MSAT at the receiving site owns that approval, and Quality Assurance at both sites countersigns it.

6.2 The analytical methods

The methods in scope are the release and characterization methods for the drug substance and the in-process methods that support the control of the intermediates. Table 6.3 lists the 9 methods to be transferred and the performance each demonstrated at qualification, from the record of qualified method performance (the method record). The two infectivity titre assays of that record (AMV-3017 and AMV-3018) are not transferred. They measure viral clearance in the virus spiking studies, which stay at small scale at the sending site (Table 8.1).

Table 6.3: Analytical methods to be transferred, with the performance each demonstrated at qualification.

Reference	Analytical method	Intermediate precision	Limit of quantitation	Accuracy (%)
AMV-3010	Released N-Glycan Map (2-AB HILIC-UPLC)	2.5 %	0.5 % glycan	99
AMV-3012	Host-Cell Protein by ELISA	9.5 %	1 ng/mg	95
AMV-3013	Charge Variants by icIEF	2 %	1 % acidic	99
AMV-3014	Residual Host-Cell DNA by qPCR	15 %	1 pg/mg	92
AMV-3015	Turbidity by Nephelometry (NTU)	4 %	0.5 NTU	98
AMV-3016	Leached Protein A by ELISA	6.5 %	0.25 ppm	97.4
AMV-3019	Protein Concentration by A280 (UV)	1.2 %	0.5 g/L	99.6
AMV-3021	Product Titre by Protein A Affinity HPLC	2 %	0.05 g/L	98.5
AMV-3221	Aggregate by SEC-HPLC (verification-qualified)	3.8 %	0.2 % HMW	99.1

Intermediate precision is a relative standard deviation.

The host cell protein ELISA (AMV-3012) supports the host cell protein clearance argument across all three chromatography steps, and every judgement of host cell protein against an intermediate acceptance criterion rests on its results. It is also the second least precise method in Table 6.3, at 9.5 %. Its transfer should therefore be completed and reviewed before the engineering batches are run.

Two records cover SEC, AMV-3011 for size variants and AMV-3221 for aggregate (Table 6.2). AMV-3221 is the one transferred, and Table 6.3 gives its qualified performance.

The product titre method by Protein A affinity HPLC (AMV-3021, under SOP-3016) measures the antibody in a clarified harvest. It is transferred because the titre sets the protein load of a capture run.

The master plan PCMP-001 Version 2.0 sets nrCE-SDS (AMV-3020, under SOP-3015) as the method for fragments, following the analytical method decision AMD-001 of 2026-03-20, and keeps SEC as the method for aggregate (PCMP-001 §7.8). The receiving site therefore needs nrCE-SDS for fragments and SEC for aggregate. The

nrCE-SDS method is transferred and requalified like the methods in Table 6.3. The method record holds no performance row for it. Its validation is AMV-3020 in Table 6.2, from which the transfer protocol takes the values to be reproduced.

Each method will be transferred under a written protocol and requalified at the receiving site before it is used to test comparability material. A transfer is successful when the receiving site reproduces the qualified intermediate precision of the method and returns accuracy within the limits the transfer protocol sets, on a common set of samples. The protocol also sets the allowed difference between the results of the two sites. If the receiving site cannot meet the qualified precision, the cause should be investigated and corrected before the exercise is repeated. A method may not be used for in-process or release testing until its transfer has passed.

Analytical Development at the sending site owns the transfer protocols and the comparison of the two data sets. Quality Control at the receiving site executes the transfers. Quality Assurance approves the outcome. The person who executes a method transfer may not approve it.

6.3 The control strategy

The control strategy is the set of controls that together assure the quality of the drug substance. In this transfer it has three layers. Parameter classification places every process parameter in a class and attaches a control to it. In-process limits bound what each intermediate may deliver to the next step. Release testing confirms the attributes of the drug substance against Table 2.2, with two exceptions. Residual DNA and viral clearance are met through the clearance that the characterization demonstrates, and no batch test measures them against their criteria.

Classification is performed under the parameter classification procedure SOP-4001 on the evidence of the characterization studies. Each report classifies the parameters of its own step, and the master report PCMR-001 consolidates the classification for the whole train. A parameter whose variability affects a quality attribute is a critical process parameter (CPP). It is a well-controlled critical process parameter (WC-CPP) where the risk of leaving the design space is low. A parameter that affects process consistency without affecting quality is a key process parameter (KPP). The remaining parameters are general process parameters (GPPs), which the batch record controls without a claim of criticality (CMC Biotech Working Group 2009). At this revision the train has 1 CPP, 20 WC-CPPs, 10 KPPs and 6 GPPs (Table 6.1).

In the characterization, each step output on which a drug substance criterion depends is judged against an intermediate acceptance criterion, where PCMP-001 Version 2.0 §7.3 defines one. The criterion is the drug substance criterion carried backward through the train, using the worst case that each downstream step showed on the platform products X-Mab, Y-Mab and Z-Mab. No margin is applied. The derivation uses no data from the studies it judges. Each step report checks the platform worst cases of its own step, and PCMP-001 §7.5 states when that check tightens or widens a criterion upstream. The master plan gives 22 such criteria across 6 steps.

Residual DNA has a drug substance criterion but no intermediate acceptance criterion (PCMP-001 §7.3). Its criterion, no more than 0.001 ng per dose, is the acceptable range that the A-Mab case study claims, and the case study notes that DNA is consistently cleared from the process (CMC Biotech Working Group 2009, 47). The criterion is expressed per dose. The dose is the weekly A-Mab dose of 10 mg/kg (CMC Biotech Working Group 2009, 26), or 700 mg for a reference adult of 70 kg, a body weight that this package assumes. The criterion is met through the clearance that the step reports show, and the master report PCMR-001 assesses the capability of the drug substance against it (PCMP-001 §7.3). It is not tested at release. A release test could not show it, because the qPCR method (AMV-3014) quantifies DNA only down to 1 pg/mg, which is 0.7 ng per dose, 700 times the criterion.

XMuLV clearance rests on three claimed steps and MVM clearance on two, because the low-pH step carries no MVM claim. Low pH disrupts the lipid envelope of an enveloped virus such as XMuLV, and a parvovirus such as MVM has no envelope. The anion exchange step binds both viruses by their surface charge in flow-through mode, and the virus filter retains both by their size. These mechanisms act on different properties of the virus particle. A particle that survives one step is therefore no more likely to survive the next, and the log reductions are added (International Council for Harmonisation 2023a). Each claimed step is held to its platform claim (PCMP-001 §7.4). For XMuLV, low-pH viral inactivation claims 6.6 log10, anion exchange 5.5 and virus filtration 4.62, a sum of 16.72 against a cumulative requirement of 16.7. For MVM, anion exchange claims 4 log10 and virus filtration 4.62, a sum of 8.62 against 8.6. Both requirements are met by the claims alone, with a margin of 0.02 log10 for each virus. No step is credited above its claim (PCMP-001 §7.4). With these margins every claimed step is therefore needed, and a step that falls short of its claim cannot be made up by the others. The margin is also smaller than the intermediate precision of the infectivity titre assays, 0.3 log10 for XMuLV (AMV-3017) and 0.32 log10 for MVM (AMV-3018). The load conditions on which each claim depends, such as the pH and the conductivity of the anion exchange load, are controlled by the batch record. The master report PCMR-001 states the cumulative clearance and compares it with the acceptance criteria in Table 2.2.

The control strategy is transferred when PCMR-001 has been approved and the classes it consolidates have been written into the master batch record. The engineering batches and the qualification batches are both run under that batch record (Table 10.1).

7 Process performance qualification strategy

Process performance qualification is the part of Stage 2 that follows this transfer (U.S. Food and Drug Administration 2011). Stage 2 also covers the qualification of the facility, equipment and utilities, which condition 3 below requires before the campaign starts. Process performance qualification demonstrates that the commercial process, run by the receiving site with the transferred methods and control strategy, reproducibly delivers drug substance of the required quality. This plan does not execute that demonstration. It states what must be in place before the campaign may start and the strategy the campaign will follow.

The campaign will manufacture 5 consecutive batches at commercial scale under the approved master batch record. Every parameter will be held at its set-point in Table 6.1, and no deliberate excursion will be run. Post-clarification turbidity is the exception. It is measured, not set, because it is the response of centrifugation (rcf) and depth filter load. The campaign will not challenge the edges of the characterized ranges. Those edges are established at small scale, and a qualification batch is not a characterization experiment. However, a campaign of this size cannot capture the variability that commercial manufacture will show over its life.

The campaign may start when the following conditions are met:

1. The characterization package is complete and PCMR-001 has been approved.
2. Every method used for in-process and release testing has been transferred and requalified at the receiving site.
3. The commercial equipment train and the utilities that serve it are qualified.
4. The master batch record carries the confirmed set-points, the confirmed ranges and the in-process controls.
5. The gaps in Section 8 are closed, or Quality Assurance has assessed and accepted in writing the residual risk of an open gap.

The qualification protocol will set the sampling plan of the campaign, with more sampling and testing than routine manufacture, as the lifecycle guidance expects (U.S. Food and Drug Administration 2011). The sampling plans of the step plans are for small-scale runs. Each intermediate for which PCMP-001 Version 2.0 §7.3 defines an intermediate acceptance criterion will be tested against it, as in the characterization. The acceptance criteria of the campaign will also be stated in the qualification protocol. The master plan PCMP-001 plans the characterization campaign that produces the evidence, and the master report PCMR-001 consolidates it. Both are approved before the qualification campaign starts.

Continued process verification begins when the qualification campaign is complete. The receiving site will draft the monitoring plan during the campaign. The plan may not be finalized until the campaign data are available, because its control limits are derived from the performance that the commercial process actually shows (U.S. Food and Drug Administration 2011).

8 Gap analysis

The gaps in Table 8.1 were identified at the approval of Version 1.0 of this plan, by comparing what the receiving site held with what the qualification campaign requires. At this revision, the action on the scale-down models compares the engineering batches with the development train, the action on the analytical methods adds nrCE-SDS and leaves out the infectivity titre assays, and the action on parameter classification runs the engineering batches under the master batch record (Section 1.1). A gap must be closed before the qualification campaign starts, or its residual risk must be assessed and accepted under the quality system (International Council for Harmonisation 2023b).

Table 8.1: Transfer gaps, their impact, the action that closes or manages each and the function that owns it.

Gap	Impact	Action that closes or manages it	Owner
Scale-down models are qualified against the development train only	Characterization conclusions reach commercial scale only through the models	Compare the engineering batches with the development-train batches on which the models were qualified, and record the comparison in the engineering batch reports	MSAT, sending site
Analytical methods are not qualified at the receiving site	Comparability results and in-process results cannot be generated or interpreted at the receiving site	Transfer and requalify every method in Table 6.3, and nrCE-SDS for fragments, under a written protocol	Analytical Development
Harvest and UF/DF carry no multivariate design	The operating ranges of these two steps rest on univariate work and platform experience	State the limitation in the harvest and UF/DF plans PCP-004 and PCP-010, and support the ranges with univariate data and prior knowledge	MSAT, sending site

Gap	Impact	Action that closes or manages it	Owner
Parameter classification is not established until PCMR-001 is approved	The master batch record cannot be finalized	Draft the master batch record after PCMR-001 is approved, and run the engineering batches under it	MSAT, receiving site
Buffer preparation moves from preparation for each experiment to central preparation and hold	Buffer hold time and stability are not covered by development data	Qualify buffer hold times at the receiving site before the engineering batches	MSAT, receiving site
Viral clearance claims are established at small scale only	The claims cannot be verified at commercial scale	Keep the modular claims from small scale, and control the load conditions they depend on through the batch record	MSAT, sending site
Operators at the receiving site have not run this process	An execution error during qualification would invalidate a batch	Train the operators on the master batch record, and run engineering batches before qualification	Manufacturing, receiving site

The first two gaps, on the scale-down models and on the analytical methods, each need laboratory work at a site that has not yet done it. The gaps on parameter classification, buffer preparation and operator training are closed by sequencing, by a bounded qualification study and by training, and none of them needs a change to the process itself. The gaps on harvest and UF/DF and on viral clearance are not closed but managed, by the stated limitation and by keeping the claims made at small scale.

The gap on harvest and UF/DF will still be open when the characterization campaign ends, because no multivariate design is planned for either step at any point in the programme. It is accepted because the risk assessment RA-001 ranked none of the parameters of either step high and expected none of them to interact with another parameter (PCMP-001 Version 2.0 §4). An interaction between two of them is therefore not expected to affect a quality attribute in Table 2.2.

9 Roles and responsibilities

The transfer is run by a joint team drawn from both sites. MSAT leads the team at each site, and Quality Assurance approves each handover. Table 9.1 gives the responsibilities that this plan assigns.

Table 9.1: Functions, sites and the responsibility this plan assigns to each.

Site	Function	Responsibility
Sending	Process Development / MSAT	Owns the process description, the scale-down models, the characterization plans and reports, and the justification of the ranges
	Analytical Development	Owns the method transfer protocols and the comparison of the data sets of the two sites
	Statistics	Owns the design and the statistical analysis of the characterization studies
Receiving	MSAT	Owns the master batch record, the equipment and utility qualification, the engineering batches and the qualification campaign
	Quality Control	Executes the transferred methods and performs in-process and release testing
	Manufacturing	Executes the engineering and qualification batches under the master batch record
Both	Quality Assurance	Approves the plans, the reports, each handover and the disposition of any open gap

A named person is assigned to each function in the roster of the transfer team, which is kept outside this plan and revised as the team changes. Depending on the size of the team at the receiving site, one person may hold more than one function. The separations of duty in Section 6.2 and Section 11 are the exception and are not waived.

10 Deliverables and schedule

Table 10.1 lists what this plan commits to and the order in which the deliverables arrive. The order is one of dependency, not a calendar. The transfer project schedule holds the planned dates and is revised as the work proceeds.

Table 10.1: Deliverables of the transfer and the dependency order in which they arrive.

Deliverable	Owner	Depends on
RA-001, the pre-characterization risk assessment	MSAT, sending site	This plan
PCMP-001, the characterization master plan	MSAT, sending site	RA-001
One characterization plan per unit operation, PCP-003 to PCP-010	MSAT, sending site	PCMP-001
Scale-down model qualification records	MSAT, sending site	The characterization plan of the step
Method transfer protocols and reports	Analytical Development	This plan
One characterization report per unit operation, PCR-003 to PCR-010	MSAT, sending site	Execution of the corresponding plan
PCMR-001, the characterization master report	MSAT, sending site	Every characterization report
Revised process description with confirmed ranges	MSAT, sending site	PCMR-001
Master batch record	MSAT, receiving site	The revised process description
Equipment and utility qualification records	MSAT, receiving site	This plan
Engineering batch reports	MSAT, receiving site	Master batch record and method transfer
Qualification protocol	MSAT, receiving site	PCMR-001 and the engineering batch reports

The characterization work sets the duration of the transfer. Neither the engineering batches nor the qualification campaign can start until PCMR-001 is approved, and PCMR-001 cannot be written until every characterization report is complete. Equipment and utility qualification and method transfer do not wait on that chain. Each

can start when this plan is approved (Table 10.1). Operator training on the master batch record follows that record and is completed before the engineering batches (Table 4.1).

This plan was approved on 2025-06-16. The risk assessment RA-001 followed on 2025-07-14 and the master plan PCMP-001 on 2025-08-04. Table 10.2 gives, for each step, the approval date of its plan and the execution window of the studies that Version 1.0 reported. A window runs from the first day of the first study of the step to the last day of its last study. Each step plan was approved before the first study of its step began. The windows of several steps overlap. The 27 studies ran from 2025-08-11 to 2026-05-15, and the Version 1.0 reports are dated 2026-07-24.

Table 10.2: Approval date of each step plan and execution window of the studies that Version 1.0 reported.

Step	Unit operation	Plan	Approved	Studies	First study start	Last study end
3	Production Bioreactor	PCP-003	2025-08-07	4	2025-08-11	2025-12-12
4	Harvest and Clarification	PCP-004	2025-10-20	1	2025-11-03	2025-11-21
5	Protein A Chromatography	PCP-005	2025-11-03	4	2025-11-05	2026-02-06
6	Low-pH Viral Inactivation	PCP-006	2026-01-05	4	2026-01-07	2026-02-13
7	Cation Exchange Chromatography	PCP-007	2026-02-02	4	2026-02-04	2026-03-27
8	Anion Exchange Chromatography	PCP-008	2026-02-02	5	2026-02-04	2026-04-17
9	Small-Virus Retentive Filtration	PCP-009	2026-04-13	3	2026-04-15	2026-05-15
10	Ultrafiltration / Diafiltration	PCP-010	2026-02-23	2	2026-02-25	2026-03-27

11 Approvals

This plan takes effect when the functions in the approval record at the front of this document have approved it. A revision is required when the scope of the transfer changes, when a unit operation is added to the train or removed from it, or when a gap in Section 8 is closed by a route other than the action recorded against it. The author of a revision may not approve it.

References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.

International Council for Harmonisation. 2004. “ICH Q5E: Comparability of Biotechnological/Biological Products Subject to Changes in Their Manufacturing Process.” ICH.

———. 2008. “ICH Q10: Pharmaceutical Quality System.” ICH.

———. 2009. “ICH Q8(R2): Pharmaceutical Development.” ICH.

———. 2012. “ICH Q11: Development and Manufacture of Drug Substances.” ICH.

———. 2023a. “ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin.” ICH.

———. 2023b. “ICH Q9(R1): Quality Risk Management.” ICH.

U.S. Food and Drug Administration. 2011. “Guidance for Industry: Process Validation — General Principles and Practices.” FDA.